FLIGHT DELAY PREDICTION: A REFINED MACHINE LEARNING APPROACH

IE500 Data Mining – Revised Project Proposal

Group 4

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Flight disruptions lead to substantial economic losses for airlines and severe inconvenience for travelers. This revised proposal outlines a data mining framework to predict domestic flight delays using historical flight and weather forecast data. To address observation sparsity across complex aviation networks, we restrict our scope to domestic flights connecting the top 15 U.S. airport hubs. We formulate the problem primarily as a binary classification task to determine whether a flight experiences a delay of more than 15 minutes, supplemented by a secondary continuous regression task predicting exact delay duration in minutes. All predictive attributes are strictly restricted to a cut-off point of $T-2$ hours prior to scheduled departure to eliminate data leakage. Using a chronological split to reflect real-world operational forecasting, we conduct a comparative evaluation between a route-average baseline, symbolic models (Decision Trees, Random Forests), and non-symbolic models (SVM, XGBoost) using F1-score, ROC-AUC, and MAE.

1 Problem Definition & Targeted Scope

Commercial aviation operates under tight scheduling constraints. Unpredicted flight delays propagate through transportation networks, causing compounding operational inefficiencies. The primary objective of this project is to build and evaluate predictive machine learning models capable of forecasting flight delays prior to departure.

To address sparsity and low observation density across nationwide routes, we narrow our scope to domestic flights operating across the **top 15 busiest U.S. airport hubs** (e.g., ATL, ORD, DFW, DEN, LAX). Focusing on this high-density network ensures sufficient sample size per route and origin-destination pair.

Prediction Cut-off Time & Leakage Prevention: To ensure realistic deployment and strictly prevent data leakage, all predictive features are restricted to information available at a fixed cut-off point of $**T-2$ hours prior to scheduled departure (`CRSDepTime`)**. Actual weather at departure or arrival is excluded in favor of predicted weather forecasts available at $T-2$ hours.

Problem Formulation: We evaluate two complementary modeling formulations:

1. **Primary Task (Binary Classification):** Let $y \in \{0, 1\}$ represent the target variable `Is_Delayed`, defined according to the standard U.S. Federal Aviation Administration (FAA) benchmark:
$$y = \begin{cases} 1 & \text{if Arrival_Delay} > 15 \text{ minutes} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$
2. **Secondary Task (Continuous Regression):** Predicting continuous arrival delay duration (Arrival_Delay $\in \mathbb{R}$) to preserve fine-grained operational severity.

2 Data Source and Feature Description

The primary dataset is derived from the U.S. Department of Transportation (DOT) Bureau of Transportation Statistics, complemented by historical weather forecasts. Filtering for the top 15 airport hubs yields a high-density dataset of approximately 50,000 flight records.

The attribute space comprises:

- **Categorical Attributes:** Operating Carrier (airline), Origin Hub, Destination Hub, Aircraft Tail Number.
- **Temporal Attributes:** Month, Day of Week, Scheduled Departure Time (`CRSDepTime`), Scheduled Arrival Time (`CRSArrTime`).
- **Continuous Numerical Attributes:** Scheduled Elapsed Time, Distance (miles).
- **Weather Forecast Attributes (at $T-2$ hours):** Predicted Wind Speed, Precipitation Probability, Visibility, and Temperature at origin/destination airports.
- **Network Lag Attributes:** Historical delay status of the incoming aircraft assigned to the flight (capturing delay propagation).

3 Data Preprocessing and Mining Methodology

Raw aviation datasets contain missing entries, high-cardinality nominal values, and varying numerical scales. Our preprocessing workflow addresses these issues systematically.

3.1 Preprocessing Pipeline

- **Missing Value Imputation:** Continuous variables (e.g., weather forecast metrics) with missing values will be imputed using median strategies via `SimpleImputer`.
- **Categorical Encoding:** Nominal variables (Carrier, Origin, Dest) will be encoded using Target Encoding or OneHotEncoder to yield numeric representations suitable for mathematical estimators.
- **Feature Normalization:** Continuous attributes (Distance, `CRSDepTime`) will be scaled using `StandardScaler`:
$$z = \frac{x - \mu}{\sigma} \tag{2}$$
This ensures algorithms such as Support Vector Machines are not skewed by differences in feature magnitudes.
- **Class Imbalance Handling:** Historical flight data typically exhibits class imbalance. We will employ Stratified Sampling and synthetic oversampling techniques (SMOTE) strictly within training splits to prevent classifier bias toward on-time flights.

3.2 Machine Learning Algorithms

In accordance with course requirements, we will evaluate symbolic and non-symbolic models against a domain-informed baseline:

- **Domain Baseline Model (Historical Route Average):** Instead of a naive majority classifier, our baseline predicts delay probability based on historical average delay rates for the specific route and time window.
- **Symbolic / Interpretable Models:**
  - *Decision Trees & Random Forests:* Built using Gini impurity criterion to extract human-readable decision rules and rank features by relative importance.
- **Non-symbolic / Advanced Models:**
  - *Support Vector Machines (SVM):* Utilizing an RBF kernel to capture non-linear decision boundaries.
  - *Gradient Boosted Trees (XGBoost):* An ensemble method leveraging regularized decision trees to optimize performance on structured tabular data.

4 Evaluation Strategy and Experimental Setup

4.1 Chronological Validation Protocol

Random cross-validation splits introduce temporal data leakage in aviation datasets. To simulate real-world deployment, we adopt a **Chronological Split (Temporal Hold-out)**:

- **Training & Tuning Set (80%):** Comprising earlier historical months, used for model fitting and hyperparameter optimization (`GridSearchCV` via Time-Series Cross-Validation).
- **Hold-out Test Set (20%):** Comprising the most recent chronological period, held out exclusively for final model evaluation.

4.2 Performance Metrics

Due to class imbalance and dual task formulation, we evaluate models using:

- **Classification Metrics:** Precision, Recall, **F1-Score** (primary optimization metric), and **ROC-AUC**.
- **Regression Metrics:** Mean Absolute Error (**MAE**) and Root Mean Squared Error (**RMSE**) for delay duration prediction.

5 Expected Results and Project Timeline

We anticipate that advanced ensemble models (XGBoost) will yield the highest F1-score and ROC-AUC performance. Furthermore, feature importance analysis from Random Forest is expected to highlight scheduled departure time and severe weather forecasts as the most influential predictors of delays.

Project Implementation Roadmap:

- **Phase 1:** Data Filtering (Top 15 Hubs), Weather Matching at $T - 2h$ & Chronological Splitting (Week 1)
- **Phase 2:** Preprocessing, Feature Engineering (Network Lag) & SMOTE Pipeline Construction (Week 2)

- **Phase 3:** Model Training, Hyperparameter Tuning & Classification vs. Regression Comparison (Week 3)
- **Phase 4:** Evaluation, Error Analysis & Final Report (Week 4)

References

[1] U.S. Department of Transportation, *"Bureau of Transportation Statistics: Airline On-Time Performance Data,"* 2023.

[2] Kaggle, *"US Flight Delays and Cancellations Dataset,"* 2020.

[3] Tan, P.-N., Steinbach, M., & Kumar, V., *"Introduction to Data Mining,"* Pearson, 2nd Edition.